

Epistatic Circuits

Walsh–Hadamard Interaction Decomposition

for Neural Network Mechanisms

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The Problem

Circuit discovery methods decompose neural networks into subgraphs that explain a behavior. Every method in current use is **additive by construction**: it scores each component independently and assembles a circuit from the top-scoring set.

Transformers are structurally non-additive. Components interact: removing one changes what another does. Additive methods cannot detect, measure, or preserve this interaction structure.

The question: what is the exact pairwise interaction structure of a circuit, what do existing methods retain of it, and does the discarded structure carry mechanism?

The Walsh–Hadamard Decomposition

A coalition sweep over n heads evaluates all 2^n subsets. The Walsh–Hadamard transform decomposes the resulting function into exact interaction orders:

- **Order 1:** individual head effects
- **Order 2:** pairwise interactions w_{ij}
- **Order 3+:** higher-order terms

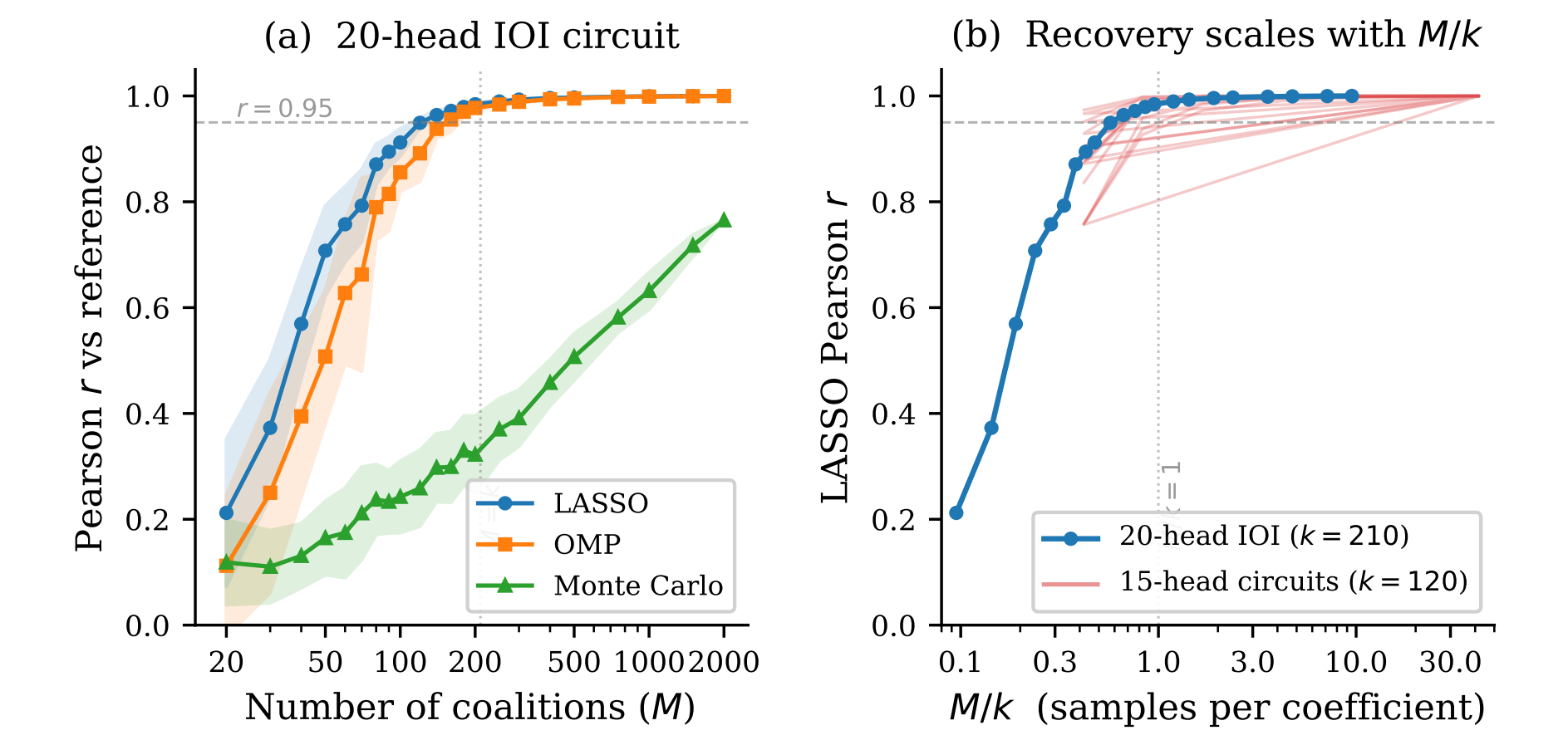
Each coefficient w_{ij} is the exact deviation from additivity for heads i and j , with no regularization and no held-out set.

Sub-additive → shared pathway
Near-zero → independent
Super-additive → mutual compensation

Compressed Sensing Makes It Scale

Brute-force cost: 2^n forward passes. For $n=20$ heads, that is $\sim 10^6$ evaluations.

The interaction spectrum is sparse. LASSO recovery from M random coalitions reaches the exact spectrum:



Correlation $r > 0.95$ at $M = 140$ for a 20-head circuit. **7,490× compression** over exact enumeration.

Theoretical brackets: generic RIP bound gives $M \approx 1,400$; structured Walsh bound gives $M \approx 13,000$. Recovery succeeds well below both.

Selective Pressures Across Methods

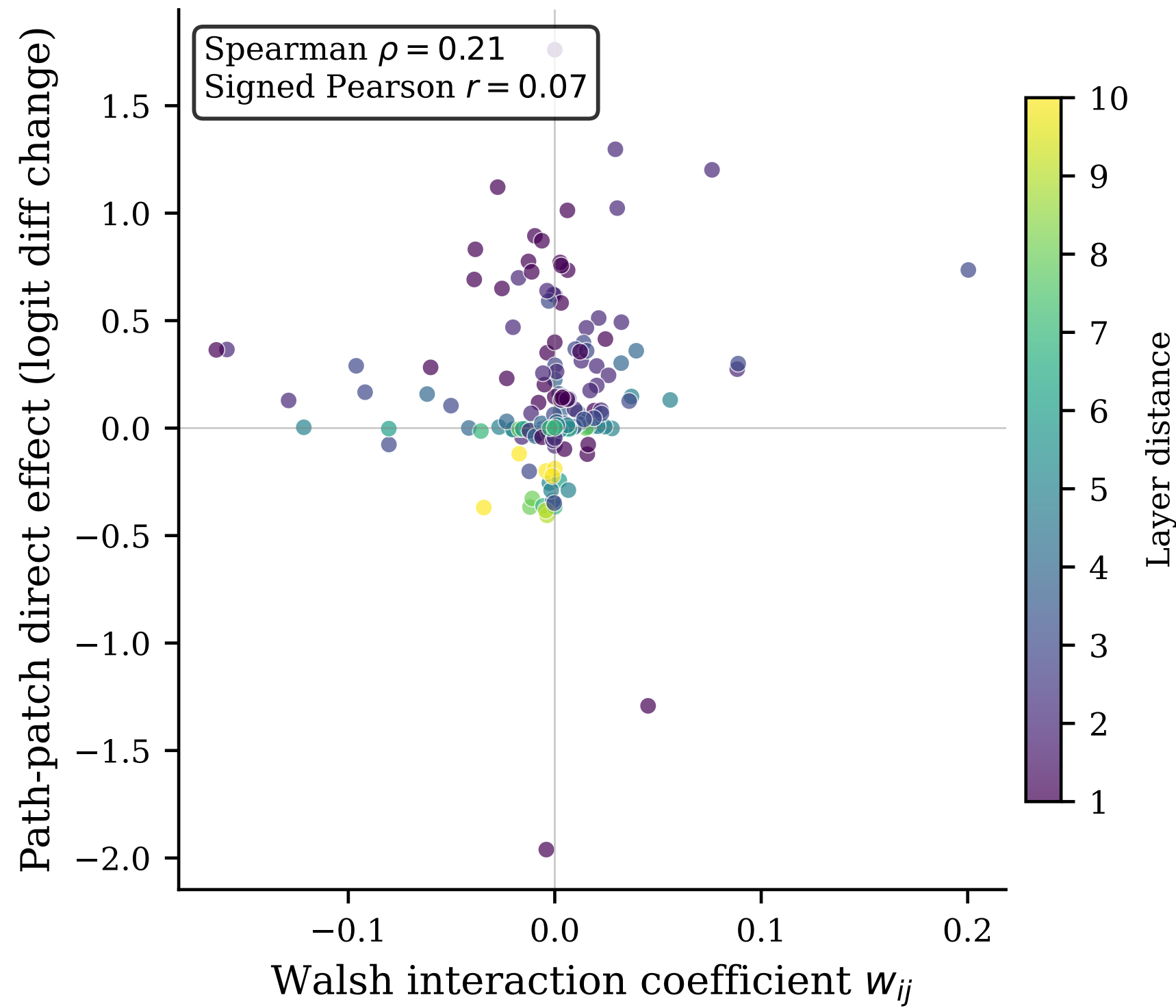
Three families of circuit proxies, audited against Walsh ground truth:

Proxy family	Agreement
Weight geometry	$R^2 < 0$ (all 30)
EAP edge attribution	$\rho = 0.58$
Path patching	$r = 0.07$

Weight geometry predicts nothing. Composition scores and subspace overlap carry zero cross-validated power for pairwise interaction under any ablation primitive.

EAP captures the linear component while missing non-linear routing effects.

Path patching is orthogonal. It measures direct causal edges; Walsh measures functional coupling. These are categorically different quantities.



The Complementation Test

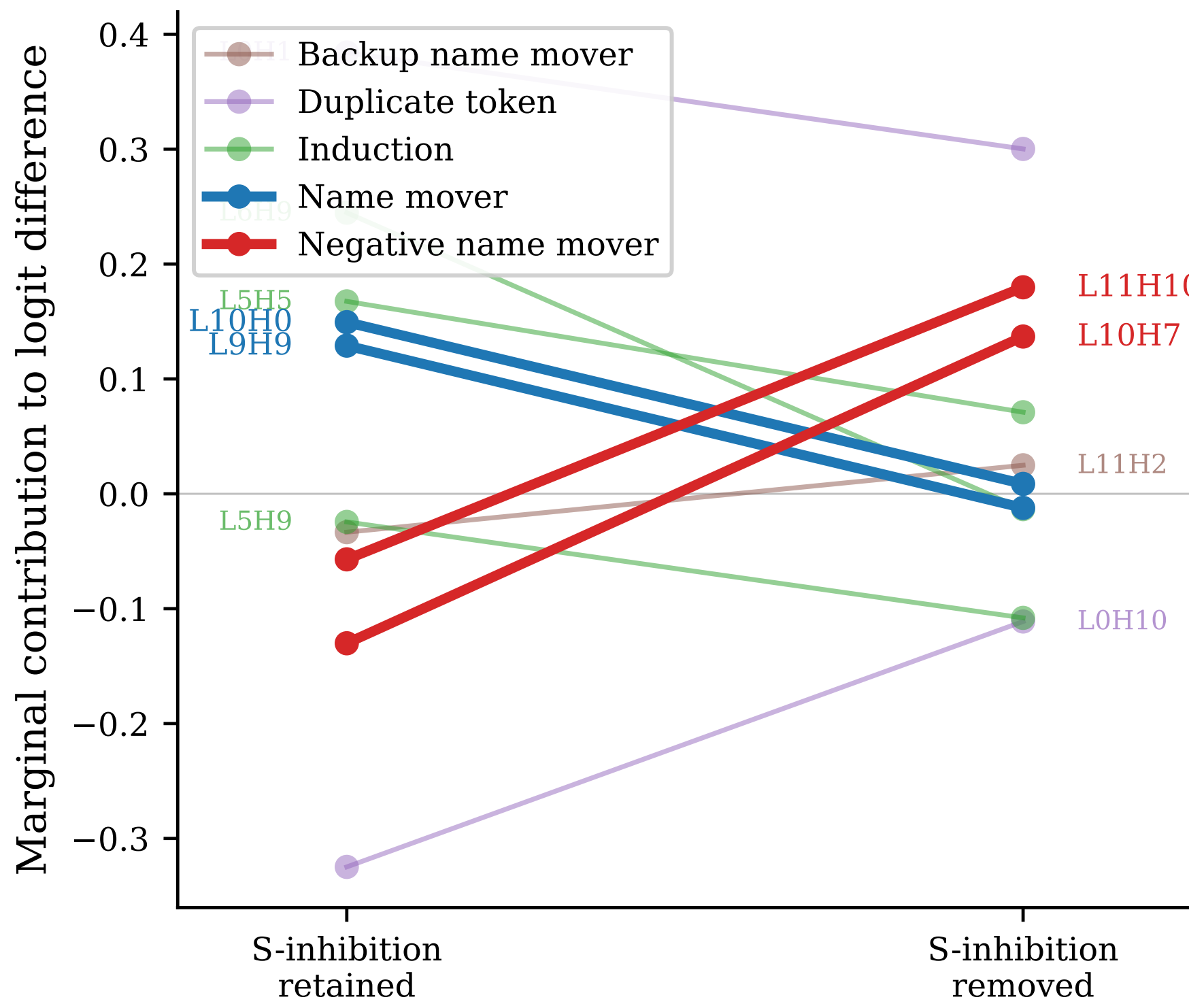
Genetics decides whether two mutations belong to one functional unit using Benzer’s cis-trans test. The Walsh interaction matrix provides exactly this data for every head pair.

Hierarchical clustering on the signed masking matrix for the 15-head IOI circuit, with cluster count selected by silhouette:

Result
ARI vs. published taxonomy
ARI vs. raw layer depth
Bootstrap merge rate (S-inh. ~ neg. NM)

S-inhibition heads and negative name movers merge into **one functional unit** in every bootstrap resample, every ablation type, and every linkage method.

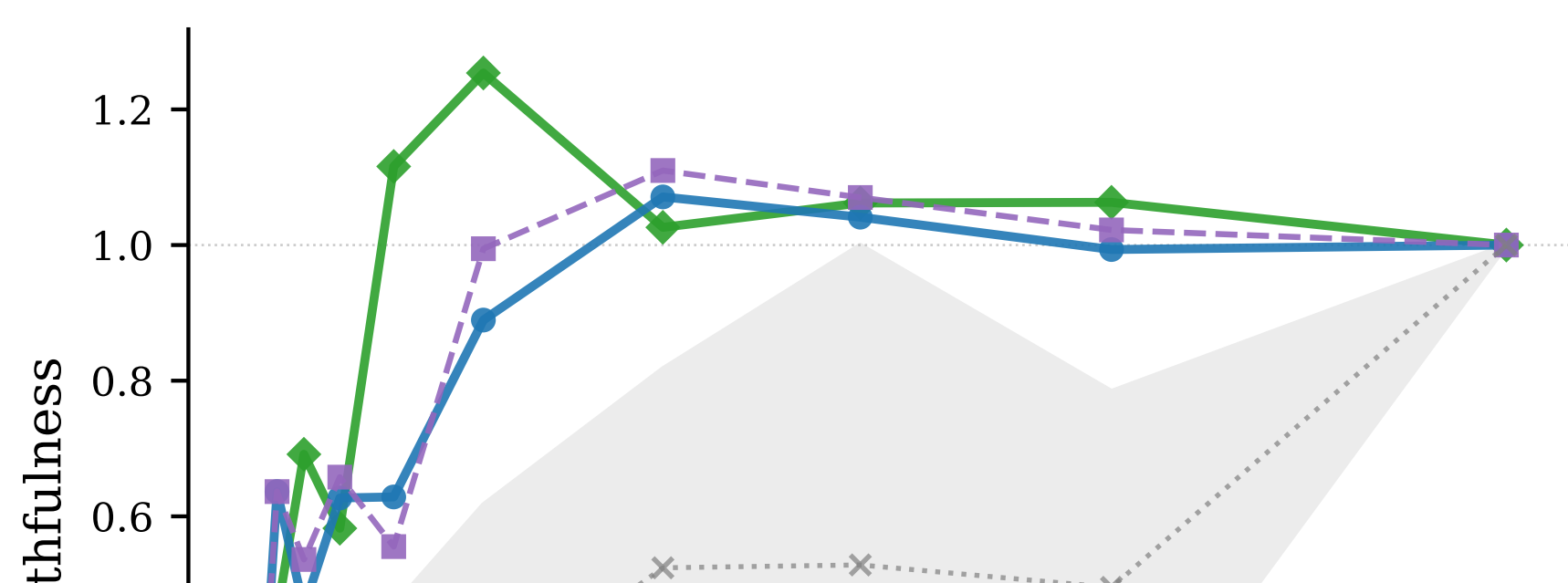
The merge is asymmetric: S-inhibition merges with negative name movers (4/4 conditions) but separates from name movers (0/2). The OV sign difference explains this — negative heads **invert** rather than diminish, so removing S-inhibition redirects their write from $-IO$ to $+IO$.



MIB Benchmark

Walsh-based circuit discovery on the MIB faithfulness benchmark:

Method	CPR
Walsh (order-1)	0.996
Activation patching	0.937
EAP-IG	0.917
ACDC	0.895



What the Discarded Structure Carries

Additive circuit summaries discard the order-2 spectrum entirely. For the IOI circuit, that spectrum carries:

- A **sign-inversion dependency** between S-inhibition and negative name movers that the original analysis recorded as unexplained
- A **functional unit boundary** that separates two head classes with identical input structure by OV sign
- A **masking hierarchy** where S-inhibition gates the function of every downstream head class
- The **super-additive** interaction ($1.9\times$) at the module level that makes backup name movers effective

None of this is visible from individual head scores. It exists in the relations between heads, which is where the Walsh

Takeaway

The pairwise interaction structure of a neural network circuit is measurable, compressible, and carries mechanism that additive circuit summaries discard.

- Walsh–Hadamard decomposition gives exact ground truth
- Compressed sensing: $r > 0.95$ from 140 coalitions (7,490× compression)
- Weight geometry predicts nothing; path patching is orthogonal; EAP captures linear coupling
- The complementation test recovers the S-inhibition / negative name-mover dependency

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4 pre-registrations with SHA commitments.